Write $P = (X, f, (a, b), E)$ for the ambient space, continuous map, endpoints, and endpoint selection. The selection has four values: none, lower, upper, both. In the following notation, regard E as its selected subset of $\{a, b\}$. The expanded pair $X_r(P)$ has components $f^{-1}[a - r, b + r]$ and $f^{-1}(\bigcup_{e \in E}[e - r, e + r])$. The shifted pair $T_t(P)$ has components $f^{-1}[a + t, b + t]$ and $f^{-1}(E + t)$. For a perturbation h, $A(P, h)$ has components $h^{-1}[a, b]$ and $h^{-1}(E)$. Pair inclusion means inclusion of both components. These results establish set inclusions; they do not assert the well-group characterization.

Result 1 (endpoint choices are exhaustive). [Topology foundations, two Boolean endpoint flags]
Let E be a selection of the lower and upper endpoints. Then

$$E = \text{none or } E = \text{lower or } E = \text{upper or } E = \text{both.}$$

Result 2 (selected endpoints lie in the interval). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let I be a pair of real endpoints. Let E be a selection of the lower and upper endpoints. Assume I.lower $\leq I$.upper (`order`). Then

$$E(I, E) \text{ is a subset of } [I.lower, I.upper].$$

Adapted from the cited source. Scope: endpoint inclusion.

Result 3 (expanded pair is nested). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let r be a real number. Assume P.interval.lower $\leq P$.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Then

$$X_r(P) \text{ is a nested pair.}$$

Adapted from the cited source. Scope: pair inclusion.

Result 4 (shifted pair is nested). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let t be a real number. Assume P.interval.lower $\leq P$.interval.upper (`order`). Then

$$T_t(P) \text{ is a nested pair.}$$

Adapted from the cited source. Scope: pair inclusion.

Result 5 (shifted pair lies in expanded pair). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let r and t be real numbers. Assume $r \geq 0$ (`radius`). Assume $-r \leq t$ and $t \leq r$ (`shift`). Then

$$T_t(P) \text{ is included in } X_r(P).$$

Adapted from the cited source. Scope: all bounded shifts.

Result 6 (perturbed pair is nested). [Bendich, Edelsbrunner, Morozov, and Patel, definition of perturbation pairs]
Let P be interlevel problem data. Let h be a continuous real-valued map on P.space. Assume P.interval.lower $\leq P$.interval.upper (`order`). Then

$$A(P, h) \text{ is a nested pair.}$$

Adapted from the cited source. Scope: pair inclusion.

Result 7 (bounded perturbation lies in expanded pair). [Bendich, Edelsbrunner, Morozov, and Patel, definition of relative well groups]
Let P be interlevel problem data. Let h be a continuous real-valued map on P.space. Let r be a real number. Assume $r \geq 0$ (`radius`). Assume h is a perturbation of P bounded by r (`bounded perturbation(P, h, r)`). Then

$$A(P, h) \text{ is included in } X_r(P).$$

Adapted from the cited source. Scope: pair inclusion.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.